

Higher-order ODEs, First-order Systems, and Peano's Existence Theorem

1 Introduction

We study the initial value problem

$$u^{(m)}(t) = f(u^{(m-1)}(t), \dots, u(t), t), \quad u^{(j)}(t_0) = \alpha_j \in \mathbb{R}, \quad 0 \leq j \leq m-1,$$

where f is continuous. Our goals are:

- 1) Convert this m -th order equation into a first-order system;
- 2) Prove Peano's existence theorem for continuous right-hand sides;
- 3) Deduce a local C^m solution to the original m -th order IVP.

2 Reduction to a first-order system

Define

$$y_1 = u, \quad y_2 = u', \quad \dots, \quad y_m = u^{(m-1)}.$$

Then

$$y'_1 = y_2, \quad y'_2 = y_3, \quad \dots, \quad y'_{m-1} = y_m, \quad y'_m = f(y_m, \dots, y_1, t).$$

Set $y = (y_1, \dots, y_m)^\top$ and $F : I \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ by

$$F(t, y) = \begin{pmatrix} y_2 \\ y_3 \\ \vdots \\ y_m \\ f(y_m, \dots, y_1, t) \end{pmatrix}.$$

The system is

$$y'(t) = F(t, y(t)), \quad y(t_0) = y_0 := (\alpha_0, \dots, \alpha_{m-1}). \quad (2.1)$$

3 Analytic prerequisites

3.1 Uniform continuity on compact sets

Lemma 3.1 (Heine–Cantor). *If $G : K \rightarrow \mathbb{R}^q$ is continuous on compact $K \subset \mathbb{R}^p$, then G is bounded and uniformly continuous.*

3.2 Equicontinuity

Definition 3.2. A family $\mathcal{F} \subset C([a, b]; \mathbb{R}^m)$ is *equicontinuous* if for all $\varepsilon > 0$ there exists $\delta > 0$ such that $|s - t| < \delta \implies \|x(s) - x(t)\| < \varepsilon$ for all $x \in \mathcal{F}$.

3.3 Arzelà–Ascoli Theorem

Theorem 3.3 (Arzelà–Ascoli). *Let $\{x_n\} \subset C([a, b]; \mathbb{R}^m)$ be uniformly bounded and equicontinuous. Then a subsequence converges uniformly to some $x \in C([a, b]; \mathbb{R}^m)$.*

4 Peano’s Existence Theorem

4.1 Peano rectangle

Choose $a, b > 0$ and define the closed rectangle

$$R = \{(t, x) \in I \times \mathbb{R}^m : |t - t_0| \leq a, \|x - x_0\| \leq b\}.$$

Assume $R \subset I \times \mathbb{R}^m$ (shrink a, b if needed). By Heine–Cantor, F is bounded and uniformly continuous on R . Set

$$M = \sup\{\|F(t, x)\| : (t, x) \in R\} < \infty, \quad h = \min\{a, b/M\},$$

with the convention $b/M = \infty$ if $M = 0$.

4.2 Euler polygonal approximations

This subsection provides a complete, line-by-line construction and analysis of the polygonal (piecewise linear) approximations used in Peano’s proof, including a corrected proof of confinement in R and an explicit inductive framing.

Step 0. Mesh and nodes

Fix $n \in \mathbb{N}$ and set the uniform step size

$$\Delta = \frac{h}{n}, \quad t_k = t_0 + k\Delta, \quad k = 0, 1, \dots, n.$$

We first construct an approximation on the right half-interval $[t_0, t_0 + h]$.

Step 1. Recursive node values and linear interpolation

Define $x_n : [t_0, t_0 + h] \rightarrow \mathbb{R}^m$ recursively by:

- (i) Initial node: $x_n(t_0) = x_0$.
- (ii) Node update (explicit Euler): $x_n(t_{k+1}) = x_n(t_k) + \Delta F(t_k, x_n(t_k)), \quad k = 0, 1, \dots, n-1.$
- (iii) Linear pieces: $x_n(t) = x_n(t_k) + (t - t_k) F(t_k, x_n(t_k)), \quad t \in [t_k, t_{k+1}].$

Thus on each $[t_k, t_{k+1}]$ the derivative is *constant*:

$$x'_n(t) = F(t_k, x_n(t_k)) \quad \text{for all } t \in (t_k, t_{k+1}).$$

Step 2. Staying inside the Peano rectangle

Lemma 4.1. *For all $k = 0, 1, \dots, n$ and all $t \in [t_k, t_{k+1}]$ (with the convention $t_{n+1} = t_n$),*

$$\|x_n(t) - x_0\| \leq M(t - t_0) \leq Mh \leq b,$$

hence $(t, x_n(t)) \in R$ and, in particular, each node $(t_k, x_n(t_k))$ lies in R so that $F(t_k, x_n(t_k))$ is well-defined.

Proof (by induction on k). Base step ($k = 0$). We have $(t_0, x_0) \in R$ by definition. For $t \in [t_0, t_1]$, linearity on $[t_0, t_1]$ gives

$$\|x_n(t) - x_0\| = (t - t_0) \|F(t_0, x_0)\| \leq (t - t_0)M \leq Mh \leq b.$$

Inductive step. Assume $(t_j, x_n(t_j)) \in R$ for all $j \leq k$. On $[t_k, t_{k+1}]$,

$$\|x_n(t) - x_n(t_k)\| = (t - t_k) \|F(t_k, x_n(t_k))\| \leq (t - t_k) M.$$

Hence, using the telescoping sum for nodes and the final partial step (now with the *precise* inequality),

$$\begin{aligned} \|x_n(t) - x_0\| &\leq \sum_{j=0}^{k-1} \|x_n(t_{j+1}) - x_n(t_j)\| + \|x_n(t) - x_n(t_k)\| \\ &\leq \sum_{j=0}^{k-1} (\Delta M) + (t - t_k)M \\ &= k\Delta M + (t - t_k)M \\ &= M(k\Delta + t - t_k) = M((t_k - t_0) + (t - t_k)) = M(t - t_0) \\ &\leq Mh \leq b. \end{aligned}$$

Thus $(t, x_n(t)) \in R$ on $[t_k, t_{k+1}]$, and in particular $(t_{k+1}, x_n(t_{k+1})) \in R$. This closes the induction. $\square$

Step 3. Uniform boundedness and equicontinuity

Lemma 4.2 (Uniform bound and Lipschitz constant). *Each x_n is Lipschitz on $[t_0, t_0 + h]$ with Lipschitz constant $\leq M$, and*

$$\sup_{t \in [t_0, t_0 + h]} \|x_n(t)\| \leq \|x_0\| + b.$$

Proof. If $s < t$ are in a single mesh interval $[t_k, t_{k+1}]$,

$$\|x_n(t) - x_n(s)\| = (t - s) \|F(t_k, x_n(t_k))\| \leq M|t - s|.$$

If $s < t$ span multiple intervals, decompose $[s, t]$ into mesh subintervals and sum. The uniform bound follows from Lemma 4.1. $\square$

Thus the family $\{x_n\}$ is uniformly bounded and equicontinuous (a common Lipschitz bound implies equicontinuity).

Step 4. Integral identity with stepwise evaluation

Define the “left endpoint selector” $\tau_n : [t_0, t_0 + h] \rightarrow [t_0, t_0 + h]$ by

$$\tau_n(s) = t_k \quad \text{whenever } s \in [t_k, t_{k+1}), \quad \tau_n(t_0 + h) = t_n.$$

Note that $\sup_s |\tau_n(s) - s| \leq \Delta$.

Lemma 4.3 (Integral form). *For every $t \in [t_0, t_0 + h]$,*

$$x_n(t) = x_0 + \int_{t_0}^t F(\tau_n(s), x_n(\tau_n(s))) ds. \tag{4.1}$$

Proof. If $t \in [t_k, t_{k+1}]$, then on each subinterval $[t_j, t_{j+1}]$,

$$x_n(t_{j+1}) - x_n(t_j) = \int_{t_j}^{t_{j+1}} F(t_j, x_n(t_j)) ds \quad (j = 0, \dots, k-1),$$

and

$$x_n(t) - x_n(t_k) = \int_{t_k}^t F(t_k, x_n(t_k)) ds.$$

Summing and using $\tau_n(s) = t_j$ for $s \in [t_j, t_{j+1})$ yields (4.1). $\square$

Step 5. Compactness via Arzelà–Ascoli

By Lemma 4.2, $\{x_n\}$ is uniformly bounded and equicontinuous on the compact interval $[t_0, t_0 + h]$. Therefore, by Arzelà–Ascoli, there exists a subsequence (not relabeled) and a function $x \in C([t_0, t_0 + h]; \mathbb{R}^m)$ such that

$$x_n \xrightarrow[n \rightarrow \infty]{\text{uniformly}} x \quad \text{on } [t_0, t_0 + h].$$

By Lemma 4.1, for all n and all t we have $(t, x_n(t)) \in R$. Passing to the limit, $(t, x(t)) \in R$ as well (since R is closed).

Step 6. Uniform continuity of F on R and a key estimate

By Heine–Cantor, F is uniformly continuous on the compact R . Hence for every $\varepsilon > 0$ there exists $\eta > 0$ such that

$$\|(t, x) - (t', x')\| < \eta \implies \|F(t, x) - F(t', x')\| < \varepsilon \quad \text{for all } (t, x), (t', x') \in R.$$

We claim that

$$\sup_{s \in [t_0, t_0 + h]} \left(|\tau_n(s) - s| + \|x_n(\tau_n(s)) - x_n(s)\| + \|x_n(s) - x(s)\| \right) \xrightarrow[n \rightarrow \infty]{} 0. \quad (4.2)$$

Indeed: $|\tau_n(s) - s| \leq \Delta \rightarrow 0$ uniformly in s ; by Lemma 4.2, $\|x_n(\tau_n(s)) - x_n(s)\| \leq M |\tau_n(s) - s| \leq M\Delta \rightarrow 0$; and $x_n \rightarrow x$ uniformly. Therefore,

$$\sup_{s \in [t_0, t_0 + h]} \|F(\tau_n(s), x_n(\tau_n(s))) - F(s, x(s))\| \xrightarrow[n \rightarrow \infty]{} 0. \quad (4.3)$$

Step 7. Passage to the limit in the integral identity

From (4.1) and (4.3), for each fixed t ,

$$\lim_{n \rightarrow \infty} x_n(t) = x_0 + \lim_{n \rightarrow \infty} \int_{t_0}^t F(\tau_n(s), x_n(\tau_n(s))) ds = x_0 + \int_{t_0}^t F(s, x(s)) ds,$$

where we used uniform convergence of the integrands and the uniform bound by M to justify passing the limit under the integral sign. Since $x_n \rightarrow x$ uniformly, we conclude

$$x(t) = x_0 + \int_{t_0}^t F(s, x(s)) ds \quad (t \in [t_0, t_0 + h]). \quad (4.4)$$

Step 8. Differentiability and the differential equation

Because $s \mapsto F(s, x(s))$ is continuous (composition of continuous functions on a compact interval), the Fundamental Theorem of Calculus implies that $x \in C^1([t_0, t_0 + h]; \mathbb{R}^m)$ and

$$x'(t) = F(t, x(t)), \quad x(t_0) = x_0.$$

Thus x is a C^1 solution of the IVP on $[t_0, t_0 + h]$.

4.3 Left half-interval via time reversal (correct construction)

To obtain a solution on the left half-interval $[t_0 - h, t_0]$, we *do not* attempt a “backward explicit Euler” with ad hoc node formulas. Instead, introduce the time-reversed variables

$$s = t_0 - t \quad (s \geq 0), \quad z(s) = x(t_0 - s).$$

Then z satisfies the forward IVP

$$z'(s) = -F(t_0 - s, z(s)) =: G(s, z(s)), \quad z(0) = x_0,$$

on $s \in [0, h]$. The map $G(s, z)$ is continuous on the compact set

$$R^- = \{(s, z) : 0 \leq s \leq h, \|z - x_0\| \leq b\},$$

and $\sup_{R^-} \|G\| \leq M$ because R^- is mapped into R by $(s, z) \mapsto (t, x) = (t_0 - s, z)$. Therefore, Steps 0–8 above apply verbatim to the forward IVP $z' = G(s, z)$ on $[0, h]$, yielding a function $z \in C^1([0, h]; \mathbb{R}^m)$ with

$$z(s) = x_0 + \int_0^s G(\sigma, z(\sigma)) d\sigma \quad \text{and} \quad z'(s) = G(s, z(s)).$$

Define x on $[t_0 - h, t_0]$ by $x(t) := z(t_0 - t)$. Then $x \in C^1([t_0 - h, t_0]; \mathbb{R}^m)$, $x(t_0) = x_0$, and by the chain rule

$$x'(t) = -z'(t_0 - t) = -G(t_0 - t, z(t_0 - t)) = F(t, x(t)).$$

Gluing this with the right-half solution from Step 8 gives a C^1 solution on the symmetric interval $[t_0 - h, t_0 + h]$.

4.4 Conclusion (Peano’s theorem for systems)

Theorem 4.4 (Peano). *If F is continuous near (t_0, x_0) , then the IVP $x' = F(t, x)$, $x(t_0) = x_0$ admits a C^1 solution on $[t_0 - h, t_0 + h]$, where $h = \min\{a, b/M\}$ as above.*

4.5 Application to the m –th order ODE

Applying the theorem to the system (2.1) for y , we obtain a C^1 solution y on $[t_0 - h, t_0 + h]$. Then $u = y_1 \in C^m$ solves the original m –th order IVP with the given initial data.

Corollary 4.5. If f is continuous near $(\alpha_{m-1}, \dots, \alpha_0, t_0)$, then the IVP

$$u^{(m)} = f(u^{(m-1)}, \dots, u, t), \quad u^{(j)}(t_0) = \alpha_j, \quad 0 \leq j \leq m-1,$$

has a C^m solution on $[t_0 - h, t_0 + h]$.

5 References (undergraduate-friendly)

- W. Rudin, *Principles of Mathematical Analysis*, Ch. 7 (uniform convergence; Arzelà–Ascoli).
- E. A. Coddington, *An Introduction to Ordinary Differential Equations*, Ch. 1–2 (Peano and Picard theorems).
- G. F. Simmons, *Differential Equations with Applications and Historical Notes*, early chapters (Peano, examples of nonuniqueness).
- W. E. Boyce and R. C. DiPrima, *Elementary Differential Equations and Boundary Value Problems*, §2 (context and examples).